

Predicting NBA Championship Contenders from Mid Season Team Snapshots Using Deep Learning

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1. Problem Statement

Every NBA season, 30 teams compete for the championship, but only a handful are true contenders. We want to answer a simple question: given how a team looks halfway through the season, can we predict how far they will go in the playoffs?

We will collect team and player statistics at the All Star break (about 50 games into the 82 game season) and use a deep neural network to classify each team into one of six outcomes: missed playoffs, first round exit, second round exit, Conference Finals, Finals appearance, or Champion.

2. Motivation

Predicting which teams will compete for the title is one of the most discussed topics in basketball. Analysts, fans, and team front offices spend countless hours debating who the real contenders are. A reliable prediction model could help front offices make smarter decisions at the trade deadline, help media generate better analysis, and give fans a deeper understanding of what separates champions from pretenders.

Most existing work on NBA prediction focuses on predicting individual game outcomes and relies on traditional machine learning methods like Random Forests and XGBoost. Very few studies have used deep learning to predict season level outcomes like championship contention. This is the gap we want to fill.

Ethical Considerations. We will make sure our model uses only performance based features (stats, records, ratings) and does not encode any biases related to player demographics. We will check whether the model unfairly favors large market teams over small market teams and report any such findings.

3. Related Works

Khanmohammadi et al. (2024) built MambaNet, a hybrid neural network combining CNNs and RNNs to predict NBA playoff game outcomes. They achieved AUC scores between 0.72 and 0.82, but their focus was on individual playoff games, not predicting championship contenders for the full season.

Zhao et al. (2023) used graph convolutional networks to model relationships between NBA teams for game prediction, reaching 71.54% accuracy. Their approach captures team interactions but does not look at player level features or roster makeup the way we plan to.

Guan et al. (2023) created NBA2Vec, which learns vector representations of NBA players from play by play data, similar to how Word2Vec learns word meanings from text. They showed these embeddings can predict playoff series outcomes, but they did not extend this idea to season level championship prediction.

4. Methods

Our approach has five main steps:

Data Collection. We will use the nba_api Python library to pull mid season team statistics (wins, losses, offensive rating, defensive rating, pace, net rating, strength of schedule) and player statistics (points, rebounds, assists, shooting percentages, player efficiency rating) for the last 20 plus NBA seasons. Each team season will be labeled with their actual playoff result.

Feature Engineering. For each team in each season, we will build a feature set that includes about 15 team level stats, individual stats for the top 8 players in the rotation (about 10 stats each), and contextual features like roster continuity and conference strength.

Model Architecture. We will build a neural network with two main parts. First, each player's stats go through a shared sub network that produces a compact player embedding (a learned representation of that player). These embeddings are combined using attention pooling, which lets the model learn which players matter most. The combined team representation is then joined with the team level stats and passed through fully connected layers to predict the playoff outcome.

Training and Evaluation. We will use cross entropy loss with class weights to handle the imbalance in our data (there is only one champion per year but many teams that miss the playoffs). We will evaluate using leave one season out cross validation so the model never sees future data. Our metrics will be accuracy, F1 score, and top 2 accuracy (did the model rank the actual champion in its top 2 picks). We will compare our model against baselines including logistic regression, Random Forest, and a plain neural network without player embeddings.

Interpretability. We will visualize the attention weights to see which players the model thinks are most important for championship contention. We will also use t-SNE to visualize the learned team embeddings and see if championship teams cluster together in the embedding space.

5. Team Members and Responsibilities

Member	Role
Shrujal Mogadati	Data collection, feature engineering, model training
Pranav Kaliaperumal	Model architecture, hyperparameter tuning, baseline comparisons
Disha Srinivasa	Evaluation, interpretability analysis, report writing

All members will collaborate on the final report and presentation. Code will be shared through a private GitHub repository.

6. Timetable

Time Period	Milestone	Who Does What
Feb 10 to Feb 24	Collect data using nba_api, clean and preprocess, exploratory data analysis	Shrujal leads data pipeline; Pranav and Disha help with exploration
Feb 24 to Mar 10	Feature engineering, implement basic MLP baseline	Pranav leads model; Shrujal builds features; Disha sets up evaluation
Mar 10 to Mar 19	Build player embedding model with attention pooling	All members work together on model development
Mar 24 to Mar 26	Spring break. Review results so far, plan next steps	Light work, regroup after break
Mar 31 to Apr 7	Train, tune hyperparameters, run baselines. Prepare progress report (due Apr 7)	Pranav leads tuning; Shrujal runs experiments; Disha writes progress report
Apr 7 to Apr 23	Run interpretability analysis (attention weights, t-SNE), refine model	Disha leads analysis; all members contribute
Apr 28 to May 7	Write final report, create visualizations	All members contribute equally
May 7 to May 15	Prepare and rehearse final presentation (due May 15)	All members present

References

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